

Forensic Video Watermarking Prototype & Evaluation

Objective

Design and experimentally evaluate a simple forensic watermarking approach for video content that remains detectable after common consumer transformations.

This is intentionally scoped as a research exercise rather than a production implementation. We are primarily evaluating your reasoning, experimental methodology, and tradeoff analysis.

Expectations

You are **not** expected to:

- produce a state-of-the-art solution
- optimize heavily for runtime/performance

We are evaluating:

- problem framing
- experimental rigor
- robustness reasoning
- adversarial thinking
- clarity of communication
- practical research judgment

Task

Design and implement a simple watermark embedding and detection pipeline for video content.

You may use any reasonable approach, including:

- spatial-domain techniques
- frequency-domain methods (DCT/DWT/etc.)
- hybrid approaches
- ML-based methods (if appropriately scoped)

You should focus on:

- forensic traceability robustness
- detector reliability
- evaluation quality

Required Evaluation

Evaluate your approach under several common transformations/attacks, such as:

- compression/transcoding
- resizing
- cropping
- additive noise
- frame dropping
- re-encoding

You may include additional attacks if desired.

Deliverables

1. Technical Report (Most Important)

~4–8 pages preferred.

Please include:

- problem framing
- assumptions and constraints
- embedding strategy
- detection strategy
- attack model
- evaluation methodology
- metrics used
- experimental results
- failure analysis
- tradeoff discussion
- future improvements

2. Reference Implementation

MATLAB or Python preferred.

The implementation should be sufficient to:

- embed watermarks
- detect/extract watermarks
- reproduce experiments

3. Experimental Results

Please provide:

- quantitative metrics
- representative examples/visualizations
- discussion of strengths and weaknesses

Time Expectation

We expect this assignment to take approximately 8–12 hours. Please prioritize clarity and rigor over completeness.

Follow-Up Discussion

After submission, we may schedule a technical discussion focused on:

- your design choices
- experimental methodology
- limitations and tradeoffs
- potential attack strategies against your approach